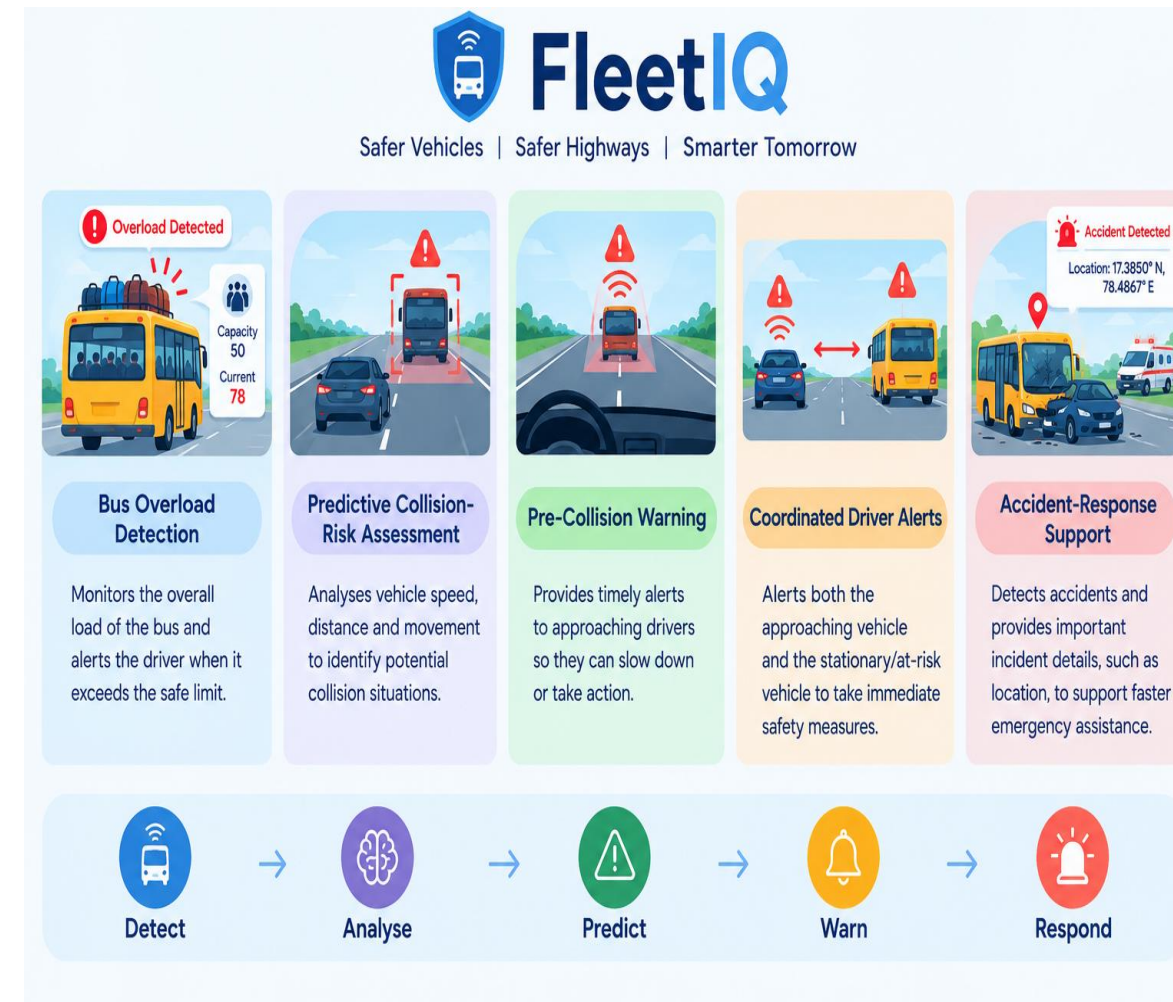




# IDEA TITLE

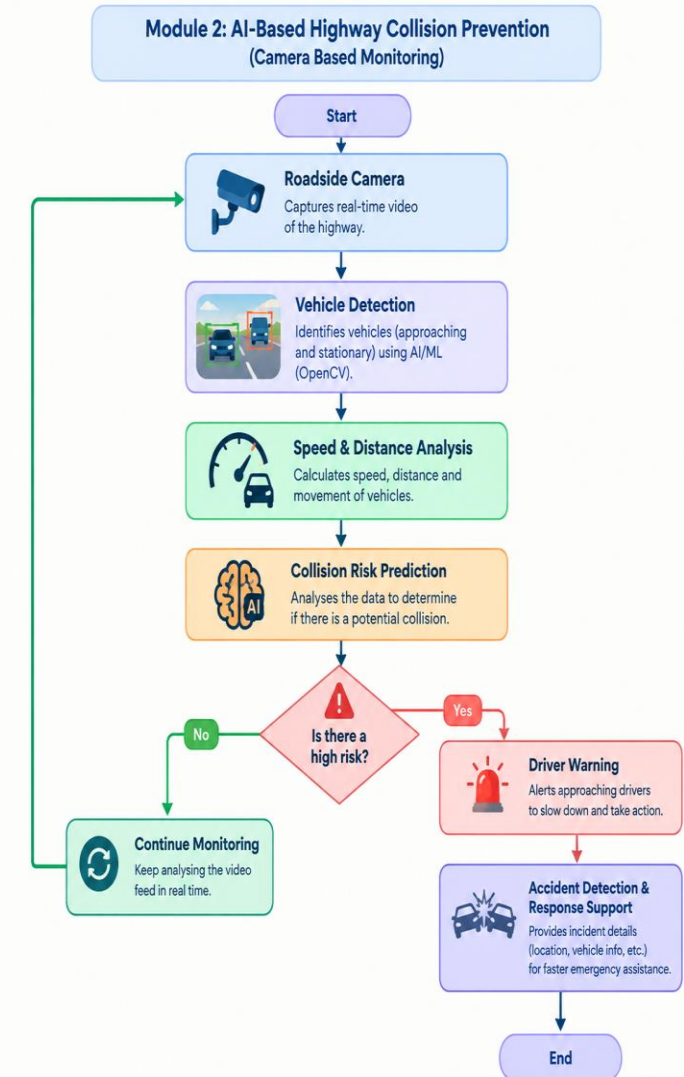
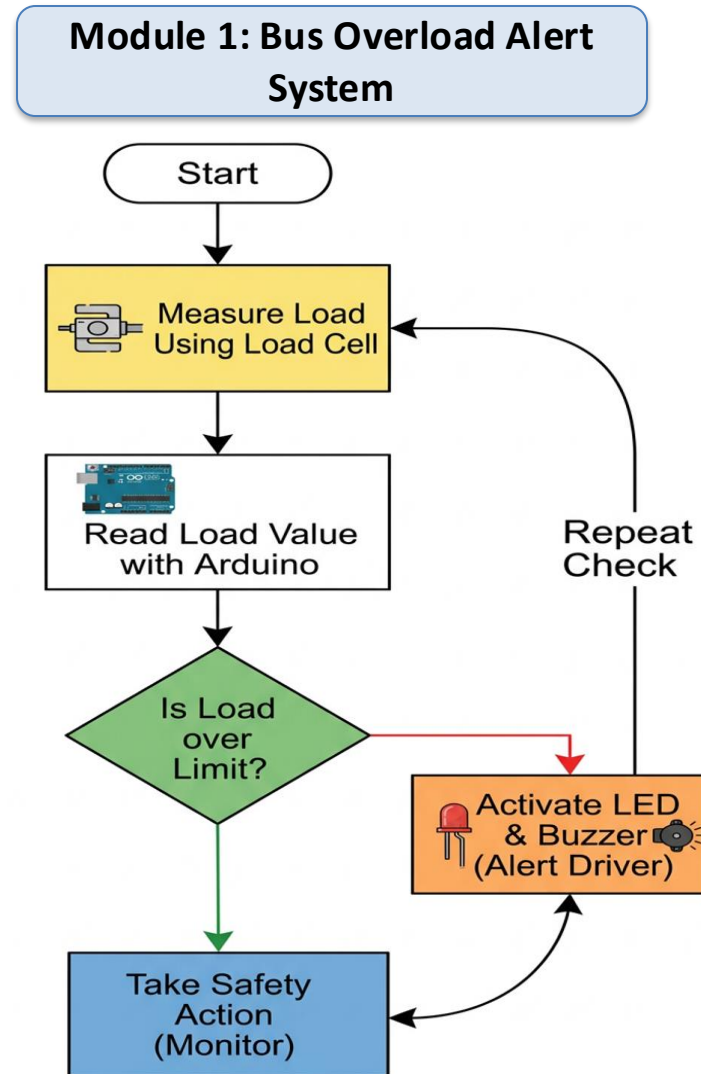
## ❖ Proposed Solution

- **Bus Overload Detection:** Monitors bus load and alerts the driver when it exceeds the safe limit.
- **Collision-Risk Prediction:** Analyses vehicle speed, distance and movement to predict possible collisions.
- **Pre-Collision Warning:** Alerts approaching drivers to slow down and avoid accidents.
- **Coordinated Alerts:** Warns both approaching and stationary vehicles about the danger.
- **Accident Response:** Provides accident location and details for faster emergency assistance.



## Technologies to be Used

- **Hardware:** Load sensors, ESP32/Arduino, buzzer, LED indicators, camera module.
- **Programming:** Python, C/C++.
- **AI & Computer Vision:** OpenCV and machine-learning models for vehicle detection and collision-risk analysis.
- **Communication & Location:** GPS and wireless communication for alerts and incident location.
- **Dashboard:** Web/mobile interface for monitoring vehicle status and alerts.



## Feasibility

- **Technical:** Uses available sensors, cameras, Arduino/ESP32 and AI-based monitoring.
- **Economic:** Uses low-cost hardware and can utilize existing highway camera infrastructure.
- **Scalable:** Can be tested on a small scale and expanded gradually.

## Strategies

- Regular sensor calibration
- Robust AI and proper camera placement
- Combine speed, distance and direction for accurate alerts
- Use local/edge processing for faster warnings
- Phased testing and continuous system improvement

## Challenges & Risks

- Sensor measurement errors
- Poor visibility and camera limitations
- False collision alerts
- Network or processing delays
- Driver response to warnings



## Impact

- **Bus Drivers:** Early overload and collision warnings.
- **Passengers:** Safer and more reliable travel.
- **Highway Users:** Reduced risk of severe collisions.
- **Emergency Services:** Faster incident information and response.

## Benefits of FleetIQ

- **Social:** Improves road safety and protects lives.
- **Economic:** Reduces accident-related losses and vehicle damage.
- **Environmental:** Helps reduce traffic disruptions and unnecessary fuel consumption.
- **Operational:** Supports safer bus operations and faster emergency response.



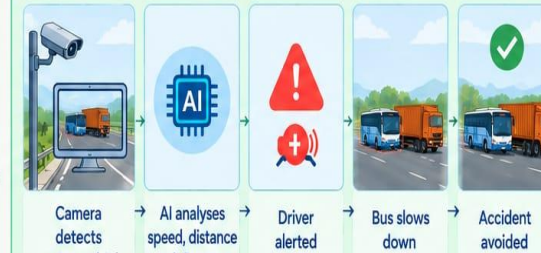
## Impact on Target Audience

Safer journeys for everyone on the road

### Before – Higher Risk



### After – Safer Journey



**FleetIQ**  
Detect • Warn • Save



### Bus Drivers

Early overload and collision warnings.



### Passengers

Safer and more reliable travel.



### Highway Users

Reduced risk of severe collisions.



### Emergency Services

Faster incident information and response.



## Benefits of FleetIQ



### Social

- Improves road safety
- Protects lives



### Economic

- Reduces accident-related losses
- Minimizes vehicle damage



### Environmental

- Less traffic disruption
- Lower unnecessary fuel consumption



### Operational

- Supports safer bus operations
- Enables faster emergency response

## Research & References

### 1. NHAI – ATMS 2023 Standards & Specifications

Studied NHAI's highway traffic management system, including *CCTV-based monitoring, accident detection and stalled-vehicle detection*.

[NHAI ATMS 2023 Document](#)

### 2. NHAI – Video Incident Detection System (VIDS)

Reviewed how *CCTV and video analytics* are used to identify incidents such as stopped vehicles and other abnormal highway conditions.

[NHAI VIDS Technical Specifications](#)

### 3. NHAI Annual Report 2023–24

Studied the implementation of *ATMS, TMCS and highway safety infrastructure* to understand existing road-monitoring capabilities.

[NHAI Annual Report 2023–24](#)

